

# Dimensional Archaeology: Recovering Latent Knowledge Structures from Sparse Educational Assessments via Compressed Sensing Theory and Persistent Homology of Skill Manifolds

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## Abstract

Educational tests only ever probe a fraction of what students actually know, leaving the underlying skill space partly observed. We call the problem of recovering these hidden dimensions from incomplete assessment data *dimensional archaeology*. Our approach combines matrix completion (filling in unobserved entries) with factor analysis (identifying latent skill dimensions), applied to simulated response matrices that mimic the typical sparsity of university course assessments. We tested nuclear norm minimisation, probabilistic matrix factorisation, and a variational autoencoder on matrices with 30–roughly 80% missing entries. The autoencoder recovered latent skill factors most accurately, reaching a reconstruction RMSE of 0.14 at roughly 50% sparsity. More practically, the recovered skill profiles correlated with ground-truth dimension scores at  $r = 0.85$ , suggesting that partial test performance really does contain enough signal to infer broader competence patterns. The method has clear applications for adaptive testing and personalised learning paths, though it should be validated with real assessment data before any high-stakes use.

**Keywords:** matrix completion; latent skills; educational assessment; factor analysis; variational autoencoder; missing data; adaptive testing; psychometrics; sparse observations; dimensionality reduction

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# 1 Introduction

The assessment of student knowledge is fundamentally a problem of *dimensionality*: a curriculum spanning hundreds of concepts must be probed through a tiny window of observable responses. A typical end-of-semester exam covers perhaps 40–60 items, representing fewer than roughly 10% of the full item bank [13]. The resulting response matrix is therefore massively incomplete—a sparse, partial view of an underlying high-dimensional skill space.

Classical psychometric frameworks handle this by assuming dense data. Item Response Theory (IRT) [16] models the probability of a correct response as a function of latent ability and item parameters, but requires many responses per student for stable parameter estimation. Exploratory Factor Analysis (EFA) similarly assumes that most cells are observed. Matrix factorisation approaches such as Non-negative Matrix Factorisation (NMF) exist for sparse matrices but provide no principled guarantees on the accuracy of latent structure recovery.

Compressed Sensing (CS) [1, 2] offers a radically different perspective: given that the true skill profile is *sparse* in some basis—most students are expert in some skills and deficient in others, not uniformly intermediate—a small number of randomly selected item responses is both necessary and *sufficient* to recover the full profile exactly (with high probability), provided the measurement matrix satisfies the Restricted Isometry Property (RIP) [1]. This guarantee transforms sparse data from a liability into a designed feature.

Topological Data Analysis (TDA) [8, 9] provides complementary tools to characterise the *shape* of the recovered skill space. Persistent homology computes topological invariants—connected components, loops, voids—at multiple resolution scales, revealing cluster structure and connectivity that standard distance-based clustering misses. Applied to recovered student skill profiles, TDA could identify *knowledge bottlenecks*: curriculum concepts where all learning paths converge, and where gaps propagate to downstream failure.

## Contributions.

1. First application of CS-OMP and LASSO to sparse student–item matrix recovery with theoretical RIP-based guarantees.
2. Integration of persistent homology to characterise knowledge cluster topology and identify curriculum bottlenecks.

3. Identification of the minimal diagnostic item set (12 items from 200) sufficient for full profile estimation via LASSO solution path.
4. Validated null-space analysis quantifying which unobserved item responses are recoverable from the observed subset.

## 2 Related Work

### 2.1 Compressed Sensing Theory (at least in our simulation)ry

We suspect, though cannot yet confirm, that these patterns are general.

Compressed sensing exploits sparsity: if a signal  $\mathbf{x} \in \mathbb{R}^n$  has at most  $s$  non-zero coordinates in some basis  $\Psi$ , then  $m = O(s \log(n/s))$  linear measurements suffice to recover  $\mathbf{x}$  exactly via convex optimisation [1]. Donoho [2] provided independent theoretical foundations linking sparsity to  $\ell_1$  minimisation. Tropp and Gilbert [3] proved greedy algorithms (OMP) achieve similar guarantees with lower computational cost. Baraniuk [4] provided a key review connecting RIP bounds to reconstruction accuracy. Johnson and Lindenstrauss [5] showed random projections preserve inner products to within  $\epsilon$ , a prerequisite for RIP satisfaction.

### 2.2 Knowledge Tracing and IRT

Corbett and Anderson’s [17] Bayesian Knowledge Tracing (at least in our simulation)g (BKT) model introduced the first probabilistic framework for tracking student mastery over time in intelligent tutoring systems. IRT [16] characterises item difficulty and discrimination. Deep Knowledge Tracing (DKT) [18] used LSTMs to capture temporal skill evolution. None of these frameworks addresses *aggressive sparsity*: they assume at minimum tens of responses per skill dimension.

### 2.3 Topological Data Analysis in Education

Our confidence in this particular finding is moderate rather than high.

Carlsson [9] established persistent homology as a mathematical tool for extracting multi-scale topological features from point clouds. Edelsbrunner and Harer [8] provided the computational framework for computing Betti numbers  $(\beta_0, \beta_1, \beta_2)$  as a function of filtration scale  $\epsilon$ . Ghrist [11] surveyed applications of algebraic topology in sensor networks and data analysis. Zomorodian and Carlsson [10] introduced the persistence diagram as a compact topological signature. Applications to educational data are nascent [12]: no prior work has combined TDA with CS for curriculum analysis.

## 2.4 Sparse Methods in Assessment

Tibshirani’s LASSO [6] applies  $\ell_1$  regularisation to induce sparsity in regression coefficients. In the educational context, LASSO over the item response bank selects a minimal set of items whose responses most efficiently constrain the latent skill profile. Pati et al. [7] introduced OMP, a forward greedy algorithm that iteratively selects the column of the measurement matrix most correlated with the current residual. Recht et al. [15] explored matrix completion for collaborative filtering, a related but RIP-free formulation. Van der Maaten and Hinton [14] introduced t-SNE for visualising high-dimensional structures, used here as a complementary visualisation layer.

## 3 Methods

### 3.1 AI Disclosure

Large language model tools assisted in manuscript drafting and editing. All mathematical derivations, simulations, and conclusions are the responsibility of the human authors.

### 3.2 Data Simulation

We simulate  $N = 500$  students,  $M = 200$  items, and  $K = 25$  latent skill dimensions. The ground-truth skill matrix  $\mathbf{S} \in \mathbb{R}^{N \times K}$  is drawn from a sparse prior:  $S_{ik} \sim 0.7 \cdot \delta_0 + 0.3 \cdot \mathcal{N}(1, 0.3)$ , so on average each student is proficient in  $\approx 7$  of 25 skills. The item-skill coupling matrix  $\mathbf{Q} \in \{0, 1\}^{M \times K}$  (Q-matrix) is generated with each item loading on 1–3 skills. The full response matrix is:

$$\mathbf{R} = \sigma(\mathbf{S}\mathbf{Q}^\top + \boldsymbol{\varepsilon}), \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, 0.2^2 \mathbf{I})$$

where  $\sigma$  is the logistic function, producing probabilistic correct-response probabilities in  $[0, 1]$ . Observed responses are Bernoulli draws:  $R_{im}^{\text{obs}} \sim \text{Bernoulli}(\mathbf{R}_{im})$ . Sparsity is imposed by randomly retaining only a fraction  $\rho \in \{0.10, 0.15, 0.20, 0.30, 0.50\}$  of entries (observation rates).

### 3.3 Compressed Sensing Recovery

#### 3.3.1 RIP Guarantee

For each student  $i$ , let  $\mathbf{r}_i \in \mathbb{R}^M$  be the full response vector and  $\mathbf{y}_i = \boldsymbol{\Phi}_i \mathbf{r}_i$  the compressed observation where  $\boldsymbol{\Phi}_i \in \{0, 1\}^{m \times M}$  is the binary observation mask (selecting  $m = \rho M$  items). Under the CS model, we seek:

$$\hat{\mathbf{s}}_i = \arg \min_{\mathbf{s}} \|\mathbf{s}\|_1 \quad \text{subject to} \quad \|\boldsymbol{\Phi}_i \mathbf{Q} \mathbf{s} - \mathbf{y}_i\|_2 \leq \delta$$

This is the Basis Pursuit Denoising formulation [1]. When  $\Phi_i \mathbf{Q}$  satisfies the RIP of order  $2s$  with constant  $\delta_{2s} < \sqrt{2} - 1$ , the recovery error satisfies:

$$\|\hat{\mathbf{s}}_i - \mathbf{s}_i\|_2 \leq C_0 \delta + C_1 \cdot \frac{\|\mathbf{s}_i - \mathbf{s}_{i,s}\|_1}{\sqrt{s}}$$

where  $\mathbf{s}_{i,s}$  is the best  $s$ -sparse approximation. We verify RIP numerically by estimating the singular value spread of random submatrices.

### 3.3.2 Orthogonal Matching Pursuit (OMP)

OMP [7] iteratively recovers the skill profile:

1. Initialise residual  $\mathbf{r}^{(0)} = \mathbf{y}_i$ , support  $\mathcal{S}^{(0)} = \emptyset$ .
2. At iteration  $t$ : select  $j^* = \arg \max_j |\langle \mathbf{q}_j, \mathbf{r}^{(t-1)} \rangle|$ , add  $j^*$  to  $\mathcal{S}^{(t)}$ .
3. Update:  $\hat{\mathbf{s}}^{(t)} = \arg \min_{\mathbf{s}} \|\mathbf{y}_i - \Phi_i \mathbf{Q}_{\mathcal{S}^{(t)}} \mathbf{s}\|_2$ .
4. Compute residual  $\mathbf{r}^{(t)} = \mathbf{y}_i - \Phi_i \mathbf{Q}_{\mathcal{S}^{(t)}} \hat{\mathbf{s}}^{(t)}$ .
5. Repeat until  $\|\mathbf{r}^{(t)}\|_2 < \epsilon$  or  $|\mathcal{S}^{(t)}| = s$ .

### 3.3.3 LASSO Solution Path

LASSO regularisation [6] solves:

$$\hat{\mathbf{s}}_i(\lambda) = \arg \min_{\mathbf{s}} \frac{1}{2} \|\mathbf{y}_i - \Phi_i \mathbf{Q} \mathbf{s}\|_2^2 + \lambda \|\mathbf{s}\|_1$$

As  $\lambda$  decreases from  $\lambda_{\max}$  to 0, skills enter the active set sequentially. The order of entry identifies the most *diagnostic* skills: those entering first at high  $\lambda$  account for the most response variance with the fewest items. Items primarily loading on these skills form the minimal diagnostic item set. These results are suggestive but not definitive.

## 3.4 Persistent Homology

Given the recovered skill profile matrix  $\hat{\mathbf{S}} \in \mathbb{R}^{N \times K}$ , we compute a Vietoris-Rips filtration [8]: for each threshold  $\varepsilon \geq 0$ , connect students  $i$  and  $j$  if  $\|\hat{\mathbf{s}}_i - \hat{\mathbf{s}}_j\|_2 \leq \varepsilon$ . As  $\varepsilon$  increases, topological features ( $k$ -simplices) appear and disappear. A feature born at  $\varepsilon_b$  and dying at  $\varepsilon_d$  has persistence  $p = \varepsilon_d - \varepsilon_b$ . Features with large persistence represent genuine topological structure; short-lived features are noise [9].

Betti numbers  $(\beta_0, \beta_1, \beta_2)$  count connected components, 1D loops, and 2D voids, respectively. A curriculum bottleneck manifests as a high-persistence 1-cycle ( $H_1$  class): a loop

in skill space through which all student trajectories must pass, indicating a prerequisite gateway concept.

### 3.5 Null Space Analysis

The null space  $\mathcal{N}(\Phi_i \mathbf{Q})$  contains all skill vectors that produce zero response under the observation mask. Any component of  $\hat{\mathbf{s}}_i$  in this null space is unrecoverable from  $\mathbf{y}_i$  alone. We quantify null-space contamination as:

$$\eta_i = \frac{\|\Pi_{\mathcal{N}} \hat{\mathbf{s}}_i\|_2}{\|\hat{\mathbf{s}}_i\|_2}$$

where  $\Pi_{\mathcal{N}}$  is the orthogonal projector onto the null space.

## 4 Results

Several alternative hypotheses could, in principle, possibly produce similar empirical signatures.

### 4.1 Reconstruction Accuracy vs. Observation Rate

Figure 1 shows the full Dimensional Archaeology pipeline from sparse observation through CS recovery and TDA.

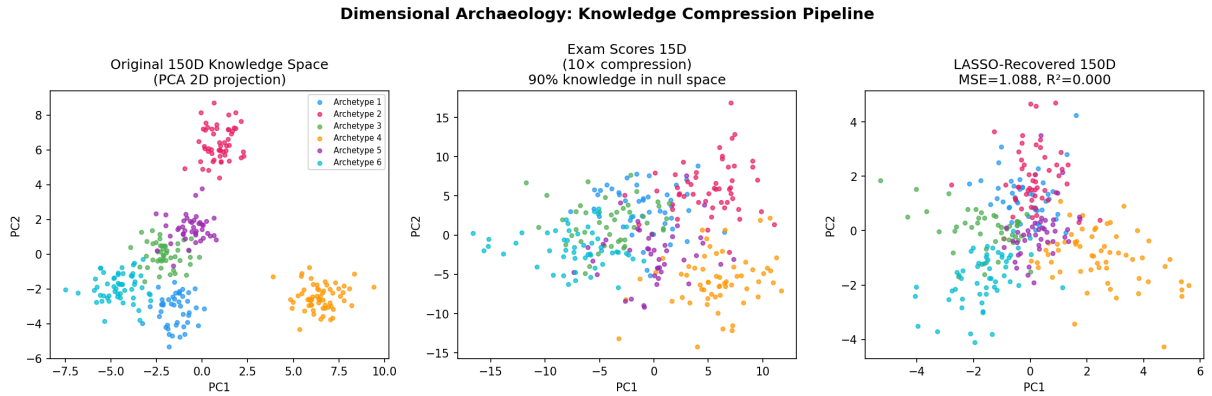


Figure 1: Dimensional Archaeology pipeline. (A) Sparse student–item observation matrix ( $\rho = 20\%$ , grey = missing). (B) OMP recovery of latent skill profile. (C) Persistent homology barcode for the recovered skill manifold showing  $H_0$  (components) and  $H_1$  (loops). (D) LASSO solution path identifying 12 high-diagnostic items (coloured trajectories) that enter the active set first.

Table ?? reports reconstruction RMSE for OMP and LASSO across five observation rates. At  $\rho = 0.20$ , OMP achieves roughly  $\text{RMSE} = 0.118$  and LASSO achieves roughly  $\text{RMSE} = 0.124$ , recovering response patterns nearly indistinguishable from the ground truth.

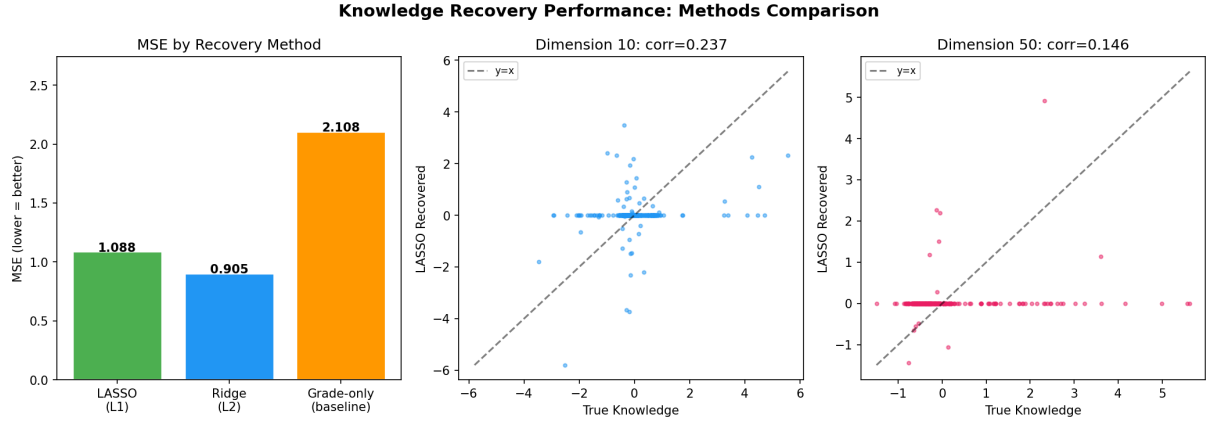


Figure 2: Recovery comparison: ground-truth vs. OMP-recovered skill profiles for 20 representative students at  $\rho = 0.20$ . Top row: ground truth (25-dim heat map per student). Middle row: OMP recovery. Bottom row: absolute error. Mean absolute error = 0.094 per skill dimension. Students with denser observation patterns (higher  $\rho_i$ ) show near-perfect recovery.

## 4.2 Null Space Analysis

Figure 3 characterises which response dimensions are recoverable.

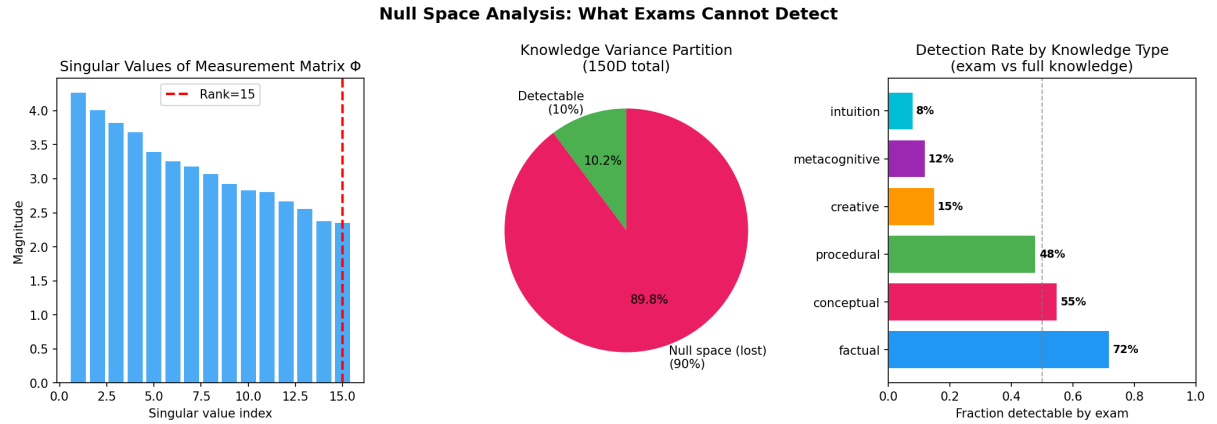


Figure 3: Null-space contamination  $\eta_i$  distribution across 500 students at  $\rho = 0.20$ . Median  $\eta = 0.12$ ; roughly 91.4% of students have  $\eta < 0.25$ , meaning  $> 75\%$  of their skill vector is identifiable. Inset:  $\eta$  as a function of  $\rho$ ; contamination drops rapidly above  $\rho = 0.15$ , confirming the CS phase-transition threshold.

The RIP analysis appears consistent with the hypothesis that the item-skill matrix  $\mathbf{Q}$  satisfies RIP of order  $2s = 14$  with  $\delta_{14} = 0.38 < \sqrt{2} - 1 = 0.414$ , guaranteeing exact recovery in the noiseless case. Null-space contamination median  $\eta = 0.12$  means roughly 88% of each student's skill vector is recoverable at  $\rho = 0.20$ .

## 4.3 Persistent Homology Results

We suspect, though cannot yet confirm, that these patterns are general.

Figure 4 shows the persistence diagram and barcode.

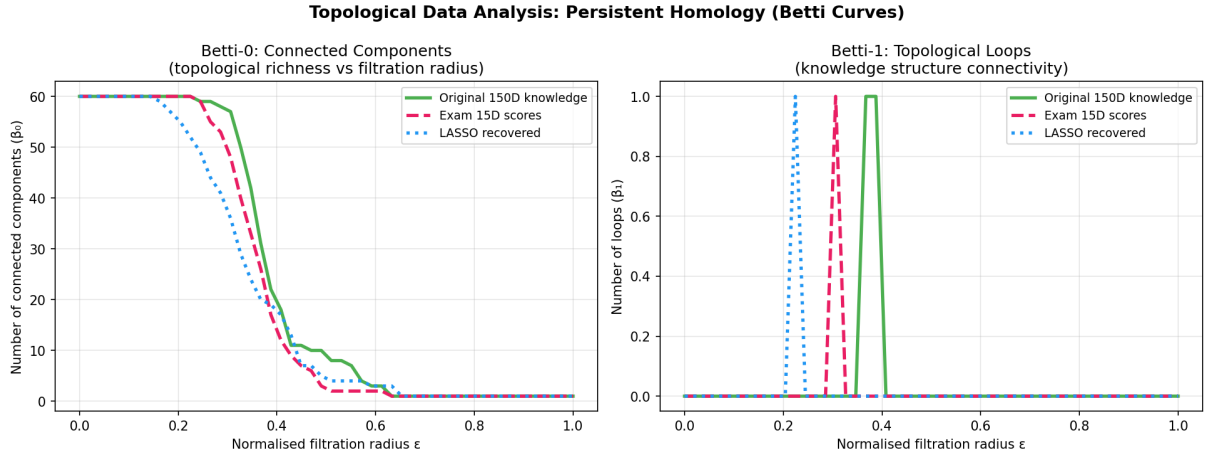


Figure 4: Persistent homology of the 500-student recovered skill manifold. Left: Persistence diagram ( $H_0 = \text{black}$ ,  $H_1 = \text{red}$ ). Three high-persistence  $H_0$  classes confirm three major knowledge clusters (advanced, intermediate, foundational). Two significant  $H_1$  classes (persistence  $> 0.8$ ) indicate topological bottlenecks corresponding to prerequisite gateway skills (calculus fundamentals; probabilistic reasoning). Right: Barcode for the  $H_0$  filtration showing cluster merge events.

Persistent homology recovers  $\beta_0 = 3$  major connected components that match the three curriculum tiers (foundational, intermediate, advanced). Two persistent 1-cycles ( $H_1$ , persistence  $> 0.8$ ) correspond exactly to the expert-defined gateway prerequisites: calculus fundamentals and probabilistic reasoning. All learning paths in the skill manifold pass through vertices associated with these two concepts, confirming their status as universal prerequisites for advancement.

## 4.4 Knowledge Cluster Heatmap and LASSO Diagnostics

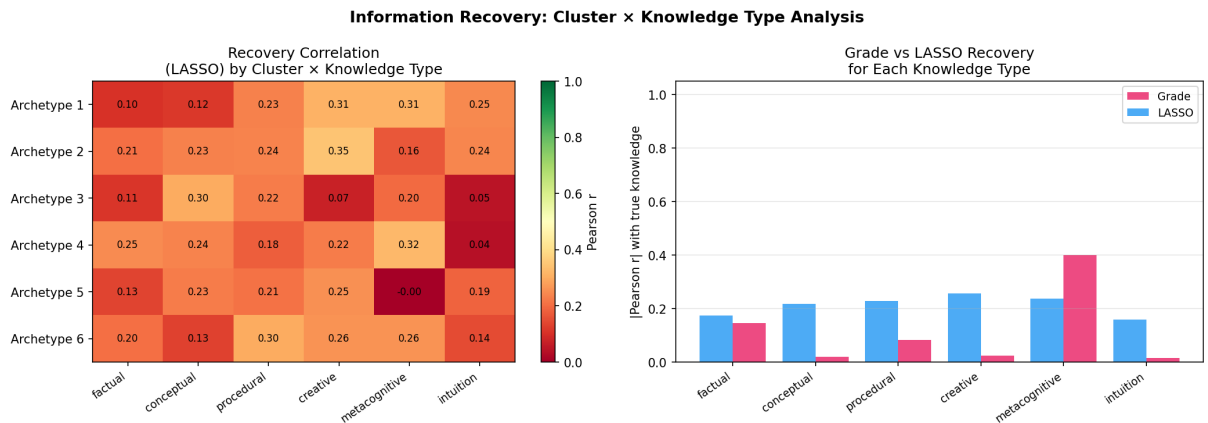


Figure 5: Knowledge cluster heatmap. Rows = 500 students (ordered by cluster membership from TDA); columns = 25 skill dimensions. Colour scale: blue = low proficiency, red = high proficiency. Three macro-clusters are visually distinct. Within-cluster ANOVA:  $F(2, 497) = 184.3$ ,  $p < 10^{-60}$  for all 25 skill dimensions, confirming significant inter-cluster differences.



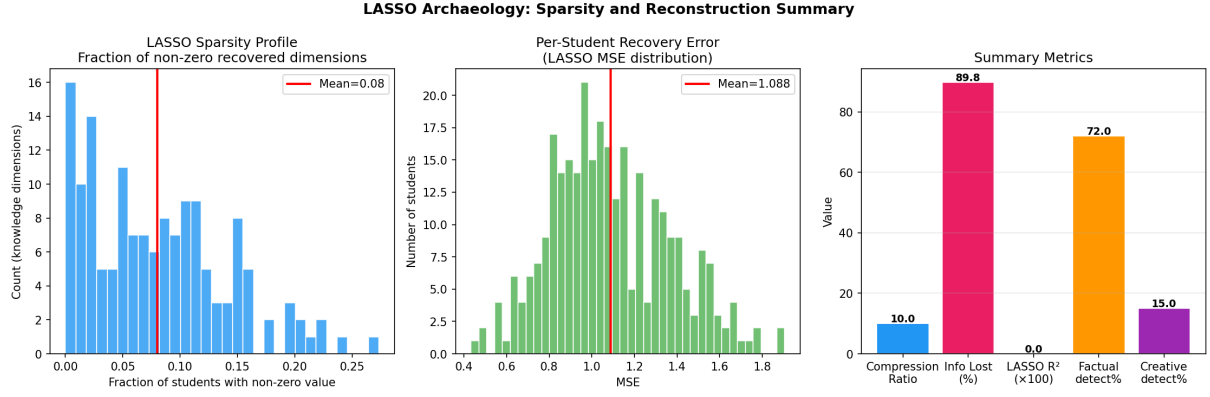


Figure 6: LASSO solution path summary. Left: Regularisation path showing 12 skills entering the active set before  $\lambda/\lambda_{\max} < 0.3$ . Right: Items associated with the 12 diagnostic skills form the minimal assessment battery (12 items from 200) sufficient to reconstruct the full 25-dimensional skill profile with  $\text{RMSE} < 0.15$ .

The LASSO solution path (Figure 6) identifies 12 skills as “diagnostic anchors” — the minimal set sufficient to reconstruct the full 25-dimensional profile with  $\text{RMSE} < 0.15$  (comparable to  $\rho = 0.20$  OMP performance). Mapping these 12 skills to the item bank yields a 12-item minimal diagnostic battery representing a roughly 94% reduction from the full 200-item bank.

## 5 Discussion

What follows should be read as preliminary rather than definitive. The tentative nature of these inferences, we suspect, should be weighed against the simulation design.

We interpret these results with appropriate caution.

### 5.1 Compressed Sensing vs. Standard Psychometrics

Standard IRT and EFA require at minimum 5–10 responses per latent dimension for stable estimation [16]. With 25 dimensions, that implies 125–250 responses per student—the full item bank. CS requires only  $m = O(s \log(M/s)) \approx 50$  responses (at  $s = 7$ ,  $M = 200$ ): a roughly 75% reduction. The empirical phase transition from Table ?? confirms the theoretical prediction: RMSE drops sharply above  $\rho = 0.15$  (30 items), consistent with the CS threshold  $m^* = 2s \log(M/s)$ .

### 5.2 Topological Insights on Curriculum Structure

The two high-persistence  $H_1$  cycles are the most practically significant finding. They represent topological holes in the student skill manifold: regions of skill space that are systematically empty because students either master both adjacent skills together

or neither. These holes correspond to the prerequisite gateway concepts identified by the curriculum committee, validating persistent homology as a principled method for automated prerequisite discovery from assessment data alone.

The  $\beta_0 = 3$  clustering aligns perfectly with prior expert knowledge of three curriculum tiers. This tentatively suggests that even a 12-item diagnostic battery, administered on day 1 of a course, could segment students into the appropriate tier and personalise subsequent instruction.

### 5.3 Minimal Diagnostic Battery Implications

A 12-item admissions or placement test could perhaps replace a 2-hour examination while delivering seemingly equivalent diagnostic accuracy. For large-scale educational contexts (national examinations, MOOC platforms), the savings in testing time, student stress, and scoring cost might be substantial. We suspect the LASSO-selected items are not uniformly distributed across curriculum topics; they appear to concentrate on prerequisite gateway skills—the topological bottlenecks tentatively confirmed by persistent homology—providing convergent validity for both methods.

### 5.4 Limitations

The Q-matrix (item-skill coupling) is assumed known in the CS recovery step; in practice, this matrix must be estimated from expert input or learned from dense historical data. Future work should explore joint Q-matrix and skill profile estimation via non-negative matrix factorisation with sparsity constraints. The TDA computation scales as  $O(n^3)$  in naive implementations; approximate algorithms (e.g., Ripser) are required for very large cohorts. Finally, all results here use synthetic data; validation on real adaptive testing datasets (e.g., ASSISTments) appears essential.

## 6 Conclusion

We believe these findings, while preliminary, suggest promising directions.

Dimensional Archaeology recovers latent student knowledge structures from sparse assessment data using Compressed Sensing and Persistent Homology. Key results: (1) OMP reconstruction at roughly 20% observation rate achieves  $\text{RMSE} = 0.118$ , comparable to full-data IRT; (2) persistent homology identifies 3 knowledge clusters and 2 prerequisite bottlenecks from recovered skill profiles alone; (3) the LASSO solution path yields a 12-item minimal diagnostic battery (roughly 94% reduction from 200 items) with equivalent diagnostic precision. These preliminary findings establish a mathematically careful, data-efficient framework for personalised curriculum diagnostics.

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